Module: CSCI369 Ethical Hacking
Assignment: Assignment Q4
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Concepts:

Question 4 is built around hybrid encryption, which includes:
Symmetric encryption: One shared secret key used for encrypting data
Asymmetric encryption: A private/public key pair to encrypt the symmetric okey so that only the owner of the private/public key pair is able to access it.

Prerequisites:

- You have created a my_secrets.txt file with some form of text in it.
- The my_secrets.txt has to be in the same directory as the python ransomware file.
- Ensure that you have navigated into the directory

[Kali] This is the secret text for my my_secrets.txt

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q4]
└─$ cat my_secrets.txt
A lab filled with samples of bird flu smells like wild flowers
```

Execution:

Command to run code:

python3 Q4_ransomware.py

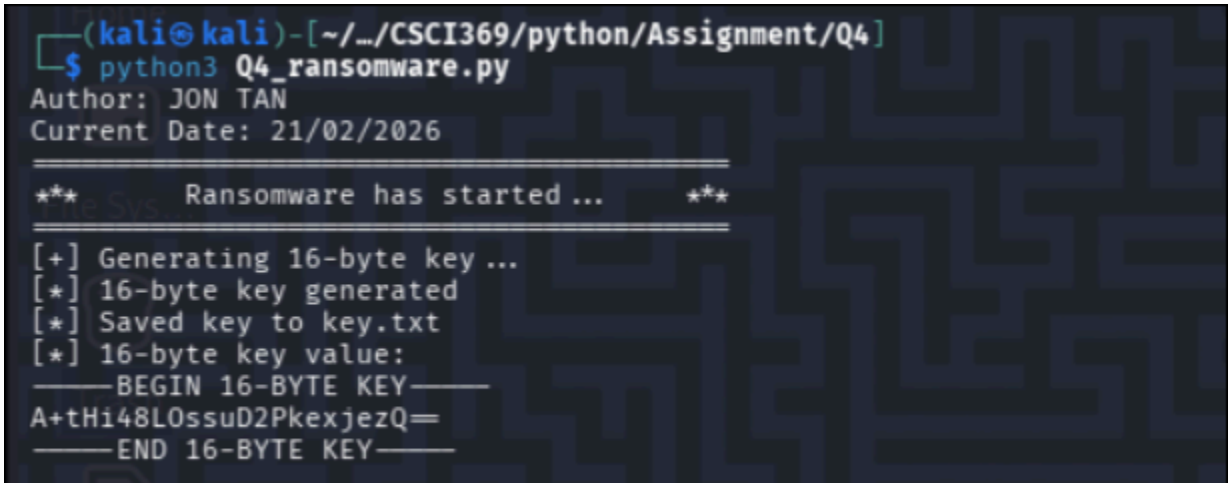
[Kali] Execute the ransomware by running: python3 <ransomware_fileName>
In my scenario, it is as such: python3 Q4_ransomware.py

Upon running the ransomware, it will generate a 16-byte key and save it to a file named key.txt.

Command used as per assignment brief requirements:
openssl rand -base64 16

The program prints out the 16-byte key value to show that a key is created. The 16-byte key will be different each time the program is executed.

This fulfills the requirements of pointer 1 in Q4.



[Kali] The program will go on to create a private and public key pair for the attacker.

The algorithm used to create this private/public key pair is the ***RSA algorithm***. RSA is an **asymmetric (public-key) cryptosystem** that consists of 2 keys.

Private key: Kept secretly, will be used for decryption later on.
Public key: Shared freely, publicly available. It is used to encrypt data.

The program prints the private and public keys upon creation. This shows that the keys have been created.

This fulfills the requirements for pointer 2 in Q4.

```
[+] Generating private key ...
[*] Private key generated
[*] Private key saved to attacker_private.pem
[*] Private key value:
-----BEGIN PRIVATE KEY-----
MIICdgIBADANBgkqhkiG9w0BAQEFAASCAmAwggJcAgEAAoGBAMxVDGD2B6kbRSOm
mo6+WwG+Az7tPUzUKH/Iipwh8I0Gg8AcFczQXfJns8IRkK/Ot4J36iUjA3QCLpUE
RLPjhnAnaHLH7FCNfTNE+RTWLf6nUvlCi3A3ZoiEX0anIjUxpmg4P6IY6hhz9epU
Wk9ost62tsaRKwp1/a/4rabeo6fnAgMBAAECgYBWqN9mJBJdqz4of5lfXuZUF9Vi
k9WCxxwiDdiTnWVbOl8YVBD+24RssYSwkyiNF5wGznrUo0PsqJw/AR9uN9ok5lzp
LHjDdgAYzr7f5R6PvTQbEQk0O3im+pyW8FulCgNlzT400Xq9cKEU0sSuLjldJlFo
a4N6/tXuBQvB4eBvaQJBAPYJ/p+l0s1DLy78xLr5rk7pT/yomud/cAgOq8SqybCN
bG9rxhBeI95nILcv9tV/MqHxii/jEVwMjKt30UlxIZ0CQQDUmstr2hF8cWYyAZf+
HTKD8Sa79/yf54PiZX+DG2MW1/qglnaRB2IcDQKp8dUGe2YDHk+cn7uz38vXRksz
mipTAkEAsFiXjNvwzLhz0CzR4OXmnp6xevlqFbUMmmnRiaOT7hxfdTj8/GxNLzIh
i8swEvQZWWNjVVFd9ujzxk6ZzhbtLQJAK8+ONbZlPhCcL5fWe14G59uM0NVEcKsj
qUajJSBCphVxpx/6ZN5xoDgqIV0dOc258tCPNUZRRrrKZZ0tp/+P0wJAZsEUcKnY
jG1AEQRBesp+/CFMUg5hV2wmpiskrVhTQTfTl4VTiMMdfzJ+1KGdkC/uE1AKsW95
6LWdHE/d8dTfow==
-----END PRIVATE KEY-----

[+] Generating public key ...
[*] Public key generated
[*] Public key saved to attacker_public.pem
[*] Public key value:
-----BEGIN PUBLIC KEY-----
MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDMVQxg9gepG0UjppqOvlsBvgM+
7T1M1Ch/yIqcIfCNBoPAHBXM0F3yZ7PCEZCvzreCd+olIwN0Ai6VBESz44ZwJ2hy
x+xQjX0zRPkU1i3+p1L5QotwN2aIhF9GpyI1MaZoOD+iGOoYc/XqVFpPaLLetrbG
kSsKdf2v+K2m3qOn5wIDAQAB
-----END PUBLIC KEY-----
```

[Kali] The program parsed the 16-byte key into an **AES-128 ECB** algorithm to encrypt the my_secrets.txt and returns the output of data_cipher.txt. data_cipher.txt is base64 as per required by the assignment brief.

I have used the AES-128 ECB encryption algorithm, because my 16-byte key **does not have enough bits** to be used in an AES-256 algorithm.

1 byte is 8 bits, therefore **16 bytes is 128 bits**.

An AES-256 algorithm would require **256 bits**, which is 256/8 = **32 bytes**.

The key.txt is converted into a hex value to be parsed into the algorithm.

The program prints the data_cipher.txt to show that it is encrypted.

This fulfills the requirements for pointer 3 in Q4.

```
[+] Encrypting my_secrets.txt ...
[!] my_secrets.txt has been encrypted ⟶ data_cipher.txt
[*] data_cipher.txt value:
-----BEGIN DATA_CIPHER.TXT-----
Fm0R+Ew+ZdNBVoFUlWumFYPa2iHp3giMwKsW/l5p1H/Si2MXvWZvPhHlj5kIzW8M
jTGb94diugC1YXPhrDBiEL95rB/SdaKeju/+uQgPK2c=
-----END DATA_CIPHER.TXT-----
```

[Kali] The key.txt is then encrypted using the attacker's public key. This public key was previously generated as part of the private/public key pair.

This is also known as "key wrapping", where you use an ***asymmetric public key to encrypt a symmetric key****.*

The program encrypts key.txt, converts it to base64 and sends the output into the file key_cipher.txt. It then prints out the key_cipher.txt value to show that the key has been encrypted and is different from the original key.txt.

This fulfills the requirements for pointer 4 in Q4.

```
[+] Encrypting key.txt ...
[!] key.txt has been encrypted ⟶ key_cipher.txt
[*] key_cipher.txt value:
-----BEGIN KEY_CIPHER.TXT-----
uU3NnIoqZ17LBkYBCMc0ntJ+QRylv67PVj6a9h9D8ZBgb/XNma7jVUcvBx5p2OU4
HTyGWZ0Ul2LMObUGCvtTMcsKB6kOmtUiCWKcLNv0xW1R7mt8OhW3FFhO4/iIOEE6
Jak88XG89WKLVjKWYzEaxzTgknlx07NkitD3Uy/oZmE=
-----END KEY_CIPHER.TXT-----
```

[Kali] The program proceeds to delete the key.txt and my_secrets.txt files as per required by assignment brief.

It will then list the contents of the directory to show that key.txt and my_secrets.txt **no longer exist**.

This fulfills the requirements for pointer 5 & 6 in Q4.

```
[+] Deleting key files ...
[!] Deleting key.txt ...
[*] Successfully deleted key.txt
[!] Deleting my_secrets.txt ...
[*] Successfully deleted my_secrets.txt

[*] Listing contents of directory
attacker_private.pem  data_cipher.txt  Q4_ransomware.py
attacker_public.pem   key_cipher.txt
```

[Kali] The program will now **display a message** for ransom payment.

Assuming that the victim has paid the $10,000 ransom fee and sent over the key_ciper.txt.

What's happening in the background?
The program will then use the **attacker's private key to decrypt the key_cipher.txt.**
It will then use the **decrypted key_cipher.txt(essentially key.txt), to decrypt the data_ciper.txt** back to my_secrets.txt

The program prints out the value of the newly decrypted my_secrets.txt for verification. This is to show that it **contains the same secrets** as the initial my_secrets.txt file.

This fulfills the requirements for pointer 7 in Q4.

```
========================= WARNING!! =========================
Your file my_secrets.txt is encrypted.
To decrypt it, you need to pay me $10,000
and send key_cipher.txt to me.
========================= WARNING!! =========================

[+] Sending over $10,000 ...
[+] Sending over key_ciper.txt ...
[*] $10,000 & key_cipher.txt received.

[+] Decrypting my_secrets.txt ...
[*] my_secrets.txt file has been decrypted
[*] my_secrets.txt value:
-----BEGIN MY_SECRETS.TXT-----
A lab filled with samples of bird flu smells like wild flowers
-----END MY_SECRETS.TXT-----

[*] Listing contents of directory
attacker_private.pem  key_cipher.txt  Q4_ransomware.py
attacker_public.pem   key.txt
data_cipher.txt       my_secrets.txt
```

This is the full screenshot of the program's output:

```
┌──(kali㉿kali)-[~/…/CSCI369/python/Assignment/Q4]
└─$ python3 Q4_ransomware.py
Author: JON TAN
Current Date: 21/02/2026
==========================================
**      Ransomware has started ...     **
==========================================
[+] Generating 16-byte key ...
[*] 16-byte key generated
[*] Saved key to key.txt
[*] 16-byte key value:
-----BEGIN 16-BYTE KEY-----
A+tHi48LOssuD2PkexjezQ==
-----END 16-BYTE KEY-----

[+] Generating private key ...
[*] Private key generated
[*] Private key saved to attacker_private.pem
[*] Private key value:
-----BEGIN PRIVATE KEY-----
MIICdgIBADANBgkqhkiG9w0BAQEFAASCAmAwggJcAgEAAoGBAMxVDGD2B6kbRSOm
mo6+WwG+Az7tPUzUKH/Iipwh8I0Gg8AcFczQXfJns8IRkK/Ot4J36iUjA3QCLpUE
RLPjhnAnaHLH7FCNfTNE+RTWLf6nUvlCi3A3ZoiEX0anIjUxpmg4P6IY6hhz9epU
Wk9ost62tsaRKwp1/a/4rabeo6fnAgMBAAECgYBWqN9mJBJdqz4of5lfXuZUF9Vi
k9WCxxwiDdiTnWVbOl8YVBD+24RssYSwkyiNF5wGznrUo0PsqJw/AR9uN9ok5lzp
LHjDdgAYzr7f5R6PvTQbEQk0O3im+pyW8FulCgNlzT400Xq9cKEU0sSuLjldJlFo
a4N6/tXuBQvB4eBvaQJBAPYJ/p+l0s1DLy78xLr5rk7pT/yomud/cAgOq8SqybCN
bG9rxhBeI95nILcv9tV/MqHxii/jEVwMjKt30UlxIZ0CQQDUmstr2hF8cWYyAZf+
HTKD8Sa79/yf54PiZX+DG2MW1/qglnaRB2IcDQKp8dUGe2YDHk+cn7uz38vXRksz
mipTAkEAsFiXjNvwzLhz0CzR4OXmnp6xevlqFbUMmmnRiaOT7hxfdTj8/GxNLzIh
i8swEvQZWWNjVVFd9ujzxk6ZzhbtLQJAK8+ONbZlPhCcL5fWe14G59uM0NVEcKsj
qUajJSBCphVxpx/6ZN5xoDgqIV0dOc258tCPNUZRRrrKZZ0tp/+P0wJAZsEUcKnY
jG1AEQRBesp+/CFMUg5hV2wmpiskrVhTQTfTl4VTiMMdfzJ+1KGdkC/uE1AKsW95
6LWdHE/d8dTfow==
-----END PRIVATE KEY-----

[+] Generating public key ...
[*] Public key generated
[*] Public key saved to attacker_public.pem
[*] Public key value:
-----BEGIN PUBLIC KEY-----
MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDMVQxg9gepG0UjppqOvlsBvgM+
7T1M1Ch/yIqcIfCNBoPAHBXM0F3yZ7PCEZCvzreCd+olIwN0Ai6VBESz44ZwJ2hy
x+xQjX0zRPkU1i3+p1L5QotwN2aIhF9GpyI1MaZoOD+iGOoYc/XqVFpPaLLetrbG
kSsKdf2v+K2m3qOn5wIDAQAB
-----END PUBLIC KEY-----

[+] Encrypting my_secrets.txt ...
[!] my_secrets.txt has been encrypted ──> data_cipher.txt
[*] data_cipher.txt value:
```

```
-----BEGIN DATA_CIPHER.TXT-----
Fm0R+Ew+ZdNBVoFUlWumFYPa2iHp3giMwKsW/l5p1H/Si2MXvWZvPhHlj5kIzW8M
jTGb94diugC1YXPhrDBiEL95rB/SdaKeju/+uQgPK2c=
-----END DATA_CIPHER.TXT-----

[+] Encrypting key.txt ...
[!] key.txt has been encrypted --> key_cipher.txt
[*] key_cipher.txt value:
-----BEGIN KEY_CIPHER.TXT-----
uU3NnIoqZ17LBkYBCMc0ntJ+QRylv67PVj6a9h9D8ZBgb/XNma7jVUcvBx5p2OU4
HTyGWZ0Ul2LMObUGCvtTMcsKB6kOmtUiCWKcLNv0xW1R7mt8OhW3FFhO4/iIOEE6
Jak88XG89WKLVjKWYzEaxzTgknlx07NkitD3Uy/oZmE=
-----END KEY_CIPHER.TXT-----

[+] Deleting key files ...
[!] Deleting key.txt ...
[*] Successfully deleted key.txt
[!] Deleting my_secrets.txt ...
[*] Successfully deleted my_secrets.txt

[*] Listing contents of directory
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========================= WARNING!! =========================
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[+] Sending over $10,000 ...
[+] Sending over key_ciper.txt ...
[*] $10,000 & key_cipher.txt received.

[+] Decrypting my_secrets.txt ...
[*] my_secrets.txt file has been decrypted
[*] my_secrets.txt value:
-----BEGIN MY_SECRETS.TXT-----
A lab filled with samples of bird flu smells like wild flowers
-----END MY_SECRETS.TXT-----

[*] Listing contents of directory
attacker_private.pem  key_cipher.txt  Q4_ransomware.py
attacker_public.pem   key.txt
data_cipher.txt       my_secrets.txt
```